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**Subject:** Mathematics | **CBSE CLASS 10 – PERIODIC TEST 07** | **Time:** 60 Minutes
**Class:** Class X (CBSE) | **Chapter 1: Real Numbers** | **MM: 30**

**Document Type:** Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

**CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:**

1. Full marks require explicit theorem/formula declaration, complete prime factorisation or contradiction steps, and boxed final answers with units.
2. In Assertion–Reason questions, truth values of both (A) and (R) as well as whether (R) validly explains (A) are explicitly established.
3. Partial marks are allocated per step in brackets: **[0.5M, 1M, 2M]**. Alternative valid methods receive full credit.

Official Evaluation Key | Maths Manthan Meerut

## SECTION A: MCQs & Assertion–Reason

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**Q1.** For any positive integer $n$, the expression $n^3 - n$ is always divisible by: [1]

(a) 6 (b) 9 (c) 12 (d) 15

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

**Correct Option: (a) 6**

Factorising $n^3 - n$:

**Formula Used:** $n^3 - n = n(n^2 - 1) = (n-1)n(n+1)$

Among any 3 consecutive integers $(n-1), n, (n+1)$, at least one is even (divisible by 2) and one is divisible by 3. Since $\text{HCF}(2,3) = 1$, the product is always divisible by $2 \times 3 = \boxed{6}$. [1M]

**Q2.** If the sum of the LCM and HCF of two numbers is 1260 and their LCM is 900 more than their HCF, then the product of the two numbers is: [1]

(a) 194400 (b) 203400 (c) 184400 (d) 190000

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

**Correct Option: (a) 194400**

Let $L = \text{LCM}$ and $H = \text{HCF}$. Given $L + H = 1260$ and $L - H = 900$. Adding gives $2L = 2160 \implies L = 1080$, so $H = 1260 - 1080 = 180$. We know:

**Formula Used:** Product of two numbers = HCF × LCM

Product $= 180 \times 1080 = \boxed{194400}$. [1M]

**Q3.** If $p$ is a prime number, then $\sqrt{p}$ is: [1]

(a) a rational number (b) a natural number
(c) an irrational number (d) an integer

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

**Correct Option: (c) an irrational number**

We know the fundamental number-theory result:

**Formula Used:** $p$ is prime $\implies \sqrt{p}$ cannot equal $\frac{a}{b}$ $(a, b \in \mathbb{Z},\ b \neq 0)$

Hence $\sqrt{p}$ is always $\boxed{\text{an irrational number}}$. [1M]

**Q4.** The number $3.24636363\ldots$ is: [1]

(a) an integer (b) a rational number
(c) an irrational number (d) a natural number

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

**Correct Option: (b) a rational number**

Notice $3.24636363\ldots = 3.24\overline{63}$ has a non-terminating repeating (recurring) decimal block 63. We know:

**Formula Used:** Every non-terminating recurring decimal represents a rational number $\frac{p}{q}$

Thus it is $\boxed{\text{a rational number}}$. [1M]

**Q5.** The number $7 \times 11 \times 13 \times 15 + 15$ is: [1]

(a) a prime number (b) a composite number
(c) an irrational number (d) neither prime nor composite

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

**Correct Option: (b) a composite number**

Factoring out 15 from both terms:

**Formula Used:** $7 \times 11 \times 13 \times 15 + 15 = 15 \times (7 \times 11 \times 13 + 1) = 15 \times 1002$

Since it decomposes into factors greater than 1, it is $\boxed{\text{a composite number}}$. [1M]

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**Q6.** **Assertion (A):** The number $7 \times 11 \times 13 + 13$ is a composite number.
**Reason (R):** A composite number has more than two distinct positive factors. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

**Correct Option: (A) Both (A) and (R) are true and (R) is the correct explanation of (A).**

We know the definition of composite numbers:

**Formula Used:** Composite $n > 1 \iff$ positive divisors of $n$ exceed two

$7 \times 11 \times 13 + 13 = 13 \times (77 + 1) = 13 \times 78$ has distinct positive factors $1, 13, 78$, etc. Thus both (A) and (R) are true and (R) explains (A). [1M]

## SECTION B: Very Short Answer Type

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**Q7.** Find the HCF of 1260 and 7344 by prime factorisation. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Prime factorising 1260 and 7344:

$$1260 = 4 \times 9 \times 5 \times 7 = 2^2 \times 3^2 \times 5 \times 7, \qquad 7344 = 16 \times 27 \times 17 = 2^4 \times 3^3 \times 17. \qquad [1M]$$

We know the prime-factorisation HCF rule:

**Formula Used:** $\mathrm{HCF}(a, b) =$ Product of smallest powers of common prime factors

Common primes are 2 and 3, with minimum exponents $2^2$ and $3^2$:

$$\mathrm{HCF}(1260, 7344) = 2^2 \times 3^2 = 4 \times 9 = \boxed{36}. \qquad [1M]$$

**Q8.** If the HCF of 65 and 117 is expressible in the form $65m - 117$, find the value of $m$. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Prime factorising 65 and 117:

$$65 = 5 \times 13, \qquad 117 = 3^2 \times 13 \implies \mathrm{HCF}(65, 117) = 13. \qquad [1M]$$

We set the given linear expression equal to $\mathrm{HCF}(65, 117)$:

**Formula Used:** $65m - 117 = \mathrm{HCF}(65, 117)$

Substituting HCF $= 13$:

$$65m - 117 = 13 \implies 65m = 13 + 117 = 130 \implies m = \frac{130}{65} = \boxed{2}. \qquad [1M]$$

**Q9.** Show that any number of the form $8^n$, where $n \in \mathbb{N}$, can never end with the digit 0. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Suppose $8^n$ ended with the digit 0 for some natural number $n \in \mathbb{N}$. We know the trailing-zero theorem:

**Formula Used:** Any integer ending with 0 is divisible by $10 = 2 \times 5 \implies$ prime 5 divides it.

However, expressing $8^n$ as a product of primes:

$$8^n = (2^3)^n = 2^{3n}. \qquad [1.5M]$$

The only prime factor of $8^n$ is 2. By the uniqueness guarantee of the Fundamental Theorem of Arithmetic, no other prime factor (specifically 5) can appear in $8^n$. Hence $\boxed{8^n \text{ can never end with the digit 0.}}$ [1.5M]

**Q10.** Show that $15^n$ cannot end with the digit 0 for any natural number $n \in \mathbb{N}$. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

We know from the Fundamental Theorem of Arithmetic that ending with 0 requires divisibility by $10 = 2 \times 5$:

**Formula Used:** Ending digit 0 $\iff$ both primes 2 and 5 occur in the prime factorisation

Factorising $15^n$ for any $n \in \mathbb{N}$:

$$15^n = (3 \times 5)^n = 3^n \times 5^n. \qquad [1.5M]$$

Here $15^n$ contains prime factors 3 and 5, but lacks prime factor 2. By the uniqueness of prime factorisation, 2 can never divide $15^n$, so $10 \nmid 15^n$. Hence $\boxed{15^n \text{ cannot end with digit 0 for any } n \in \mathbb{N}.}$ [1.5M]

**Q11.** Prove that $\sqrt{2}$ is an irrational number. Hence, deduce that $\dfrac{5-3\sqrt{2}}{4}$ is an irrational number. **[5]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

**Part 1 (3 Marks) – Proving $\sqrt{2}$ is irrational:**

Suppose, on the contrary, that $\sqrt{2} = \frac{a}{b}$ where $a, b \in \mathbb{Z}$, $b \neq 0$, and $\text{HCF}(a, b) = 1$.

Squaring both sides: $a^2 = 2b^2$ — (1) [1M]

We know the Fundamental Theorem of Arithmetic prime divisibility lemma:

**Formula Used:** If a prime $p$ divides $a^2$, then $p$ divides $a$.

Since 2 divides $a^2$, 2 divides $a$, so $a = 2c$ for some integer $c$. Substituting in (1):

$$(2c)^2 = 2b^2 \implies b^2 = 2c^2 \implies 2 \text{ divides } b^2 \implies 2 \text{ divides } b. \quad \text{[1M]}$$

Thus 2 is a common factor of $a$ and $b$, contradicting $\text{HCF}(a, b) = 1$. Hence $\boxed{\sqrt{2} \text{ is irrational.}}$ [1M]

**Part 2 (2 Marks) – Deducing $\dfrac{5-3\sqrt{2}}{4}$ is irrational:**

Suppose, to the contrary, that $\dfrac{5-3\sqrt{2}}{4} = \dfrac{u}{v}$ is rational ($u, v \in \mathbb{Z}$, $v \neq 0$). Cross-multiplying and isolating $\sqrt{2}$:

$$5 - 3\sqrt{2} = \frac{4u}{v} \implies 3\sqrt{2} = 5 - \frac{4u}{v} = \frac{5v - 4u}{v} \implies \sqrt{2} = \frac{5v - 4u}{3v}. \quad \text{[1M]}$$

Since $u, v \in \mathbb{Z}$ and $3v \neq 0$, $\frac{5v-4u}{3v} \in \mathbb{Q}$, contradicting Part 1 that $\sqrt{2}$ is irrational. Thus $\boxed{\dfrac{5-3\sqrt{2}}{4} \text{ is irrational.}}$ [1M]

**Q12.** Prove that $\sqrt{3}$ is an irrational number. Hence, prove that $\dfrac{1}{2+\sqrt{3}}$ is an irrational number. **[5]**

**▶ STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

**Part 1 (3 Marks) – Proving $\sqrt{3}$ is irrational:**

Suppose, on the contrary, that $\sqrt{3} = \frac{a}{b}$ where $a, b \in \mathbb{Z}$, $b \neq 0$, and $\text{HCF}(a, b) = 1$.

Squaring both sides: $a^2 = 3b^2$ — (1) **[1M]**

We know the Fundamental Theorem of Arithmetic prime divisibility lemma:

**Formula Used:** If a prime $p$ divides $a^2$, then $p$ divides $a$.

Since 3 divides $a^2$, 3 divides $a$, so $a = 3c$ for some integer $c$. Substituting in (1):

$$(3c)^2 = 3b^2 \implies b^2 = 3c^2 \implies 3 \text{ divides } b^2 \implies 3 \text{ divides } b. \quad \textbf{[1M]}$$

Thus 3 is a common factor of $a$ and $b$, contradicting $\text{HCF}(a, b) = 1$. Hence $\boxed{\sqrt{3} \text{ is irrational.}}$ **[1M]**

**Part 2 (2 Marks) – Proving $\dfrac{1}{2+\sqrt{3}}$ is irrational:**

Rationalising the denominator step by step:

$$\frac{1}{2+\sqrt{3}} = \frac{1}{2+\sqrt{3}} \times \frac{2-\sqrt{3}}{2-\sqrt{3}} = \frac{2-\sqrt{3}}{2^2-(\sqrt{3})^2} = \frac{2-\sqrt{3}}{4-3} = 2-\sqrt{3}. \quad \textbf{[1M]}$$

If $2 - \sqrt{3} = r \in \mathbb{Q}$, then $\sqrt{3} = 2 - r \in \mathbb{Q}$ (as rationals are closed under subtraction), contradicting Part 1 ($\sqrt{3}$ is irrational). Therefore $\boxed{\dfrac{1}{2+\sqrt{3}} \text{ is irrational.}}$ **[1M]**

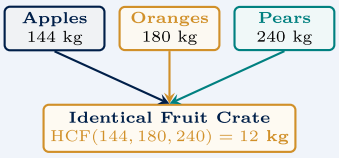
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## Q13. CASE STUDY 07: AGRICULTURAL FRUIT ORCHARD HARVEST CRATE PACKING [4 Marks]

A commercial fruit orchard in Himachal Pradesh harvests premium quality seasonal fruits. The daily harvest yields 144 kg of Royal Delicious apples, 180 kg of Kinnow oranges, and 240 kg of Bartlett pears. To minimize transport bruising, all fruits must be packed in identical wooden crates of maximum possible capacity such that each crate contains exactly one variety of fruit and no fruit is mixed.

(i) Determine the maximum weight capacity of each crate. [2]
(ii) Calculate the total number of wooden crates required to pack the entire harvest. [1]
(iii) Find the value of HCF$(144, 180) \times$ LCM$(144, 180)$. [1]

### ▶ CASE-BASED SOLUTION & MARKING SCHEME: [Total: 4 Marks]




**Harvest Logistics:** Each crate holds the greatest common factor of $144, 180$, and $240$, giving 12 kg uniform unmixed crates across $12 + 15 + 20 = 47$ crates.

We know the HCF crate-sizing formula and two-number HCF–LCM product law:

**Formula Used:** Max crate capacity $=$ HCF$(144, 180, 240)$, HCF$(a, b) \times$ LCM$(a, b) = a \times b$

**(i) Max crate capacity:** Factorising $144, 180$, and $240$:

$$144 = 2^4 \times 3^2, \qquad 180 = 2^2 \times 3^2 \times 5, \qquad 240 = 2^4 \times 3 \times 5.$$

Taking smallest exponents of common primes 2 and 3:

$$\text{HCF}(144, 180, 240) = 2^2 \times 3^1 = 4 \times 3 = \boxed{12 \text{ kg per crate}}. \qquad [2M]$$

**(ii) Total crates needed:**

$$\text{Total crates} = \frac{144}{12} + \frac{180}{12} + \frac{240}{12} = 12 + 15 + 20 = \boxed{47 \text{ wooden crates}}. \qquad [1M]$$

**(iii) Value of HCF$(144, 180) \times$ LCM$(144, 180)$:** Directly using the product theorem $(36 \times 720)$:

$$\text{HCF}(144, 180) \times \text{LCM}(144, 180) = 144 \times 180 = \boxed{25920}. \qquad [1M]$$

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